Supplementary Material

A process-based, automated SWAT$^+$/MODFLOW 6 framework for PFAS fate and transport in coupled surface water and groundwater

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This Supplement provides the full calibration and validation evidence for both legs of the coupled model and for the coupling itself. Section S1 documents the SWAT$^+$ surface-water PFAS leg: the per-HRU soil partitioning, the in-stream longitudinal profile along the mainstem, the per-reach residuals over all twenty gauged reaches, the model's full network-wide concentration field, and the predictive uncertainty ensemble. Section S2 documents the MODFLOW 6 groundwater leg: the head calibration and its spatial parameterization (pilot points and head observations), and the disaggregated PFAS plume validation that separates *prescribed* source cells from genuinely *predicted* cells. Section S3 documents the coupling — the groundwater-discharged PFAS routed into the stream network — and the joint calibration that fits the two legs against the same mainstem observations. Throughout, the groundwater figures keep the prescribed source cells visually and statistically distinct from the predicted cells, so the validation statistics are never inflated by cells whose concentration was set to the data.

S1 Surface-water PFAS calibration and uncertainty

The surface-water leg solves a soil three-phase equilibrium (aqueous, Freundlich solid, Langmuir air–water interface) per hydrologic response unit (HRU) and routes the resulting PFAS through the channel network. It is calibrated with a single global soil-loading scale factor and screened with a 120-member Latin-hypercube uncertainty ensemble. We report the all-network skill alongside the source-bearing mainstem subset: across all 20 gauged reaches the calibrated log-space RMSE is 0.74 dex with a *negative* log-space Nash–Sutcliffe efficiency (-3.7), reflecting limited skill off the mainstem where tributary soil heterogeneity dominates; along the 7 source-bearing mainstem reaches the log-RMSE is 0.15 dex. The full per-reach table is Table S1.

S2 Groundwater head calibration and PFAS validation

The same automated pipeline that builds the surface-water model generates a complete MODFLOW 6 groundwater flow-and-transport model: the SWAT$^+$ digital elevation model as the land surface, aquifer conductivity kriged from a raw well database onto 243 pilot points, recharge from the SWAT$^+$ water balance, and a streamflow-routing (SFR) network

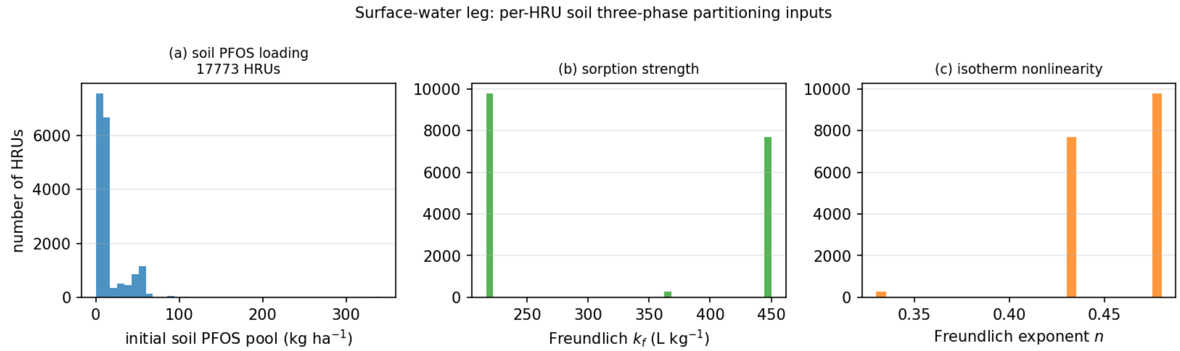

Figure S1: Distributions of the per-HRU soil three-phase partitioning inputs (the same fields mapped per HRU in the main text), shown as histograms over all HRUs: (a) the initial soil PFOS pool is right-skewed — most of the basin carries a low ambient loading with a developed-land tail; (b,c) the Freundlich sorption strength k_f and exponent n take a small number of land-use-keyed values. These fields set how strongly PFAS is held in the solid phase versus released to soil water and thence to runoff and percolation, and they are the spatial template the global soil-loading multiplier then scales.

of 1,506 reaches built from the SWAT$^+$ channel graph. A pilot-point inversion with a baseflow constraint (PEST++) calibrates the conductivity field to 5,383 per-cell water-table observations.

On this calibrated flow field, the Freundlich groundwater-transport (GWT) model is run with its source *prescribed* to the measured House Street plume — constant-concentration cells set to the observed values $> 10^4$ ng L^{-1} — not to a fitted concentration multiplier. The skill assessment therefore separates the prescribed source cells from the genuinely predicted cells.

S3 Coupling and joint calibration

The groundwater-discharged PFAS is routed downstream through the channel network by the streamflow-transport (SFT) package, closing the surface-water/groundwater mass balance on a single watershed. The two legs are then reconciled by a joint calibration against the mainstem observations.

In-stream concentration at a mainstem reach is linear in *both* the surface soil-loading and the groundwater source strength, so the combined prediction at reach i is $C_i = (L/L_0)\, C_i^{\mathrm{SW}} + g\, B_i$, where C_i^{SW} is the surface-water model's in-stream PFOS at the reference loading $L_0 = 0.11$, B_i is the cumulative groundwater-discharged load routed to reach i divided by its streamflow, and (L, g) are fitted together by non-negative least squares. Fitting the two scale factors jointly — so the surface and groundwater legs constrain each other against the same data — removes the double-counting that a naive additive sum incurs (the surface calibration, tuned to grab samples, already integrates groundwater discharge).

The joint fit credits a groundwater term that is statistically supported ($F_{1,5} = 16.2$, $p = 0.01$; $g = 0.061$ with 95% CI [0.022, 0.100] excluding zero), concentrated at the lower

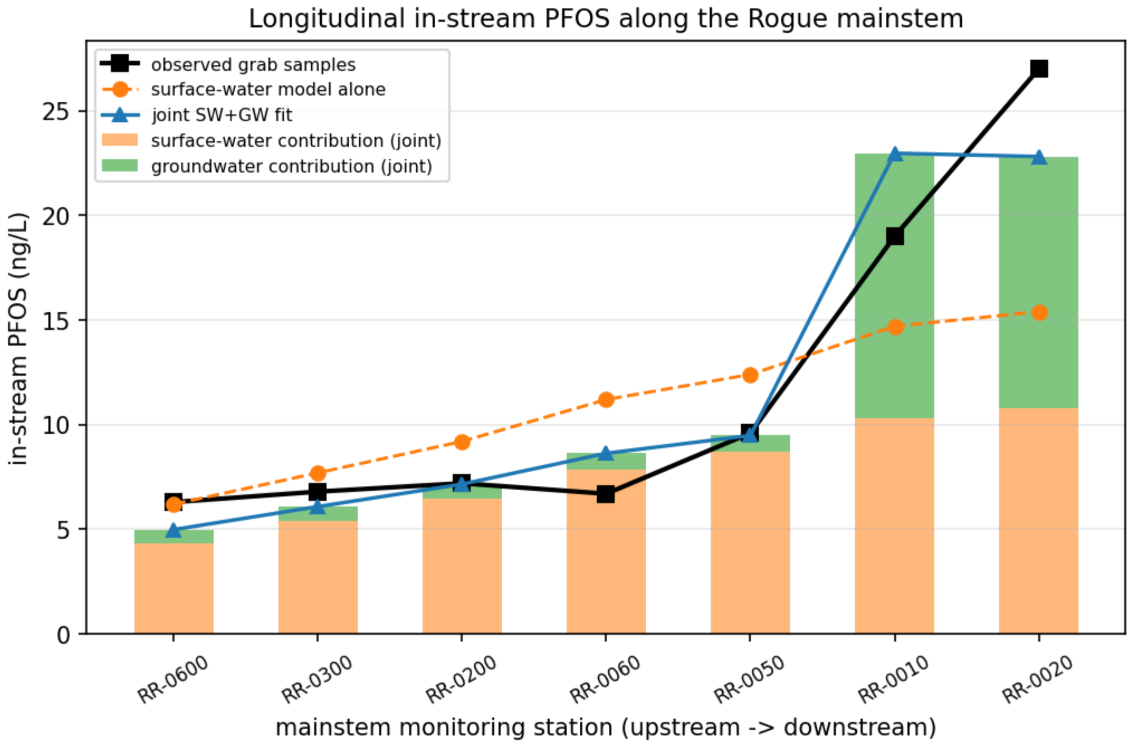

Figure S2: In-stream PFOS along the Rogue mainstem, ordered from the upstream headwater station (RR-0600) to the lowest reach near the Wolverine corridor (RR-0020). The observed grab samples (black) rise sharply at the lower mainstem from ∼7 to 27 ng L^{-1}; the surface-water model alone (orange) reproduces the upstream concentrations but underpredicts the lower-mainstem rise. The stacked bars decompose the joint surface-plus-groundwater fit into its surface-derived (orange) and groundwater-derived (green) parts: the groundwater contribution is negligible in the headwaters and becomes the larger share at the two lowest reaches, which is where it closes the residual the surface model leaves open.

mainstem near the Wolverine source, and the fitted surface soil-loading *decreases* ($L = 0.077$ versus 0.11) — the signature of a resolved double-counting. With only seven source-bearing reaches the magnitude is poorly constrained, and the joint fit alone cannot distinguish groundwater discharge from any unmodeled term co-located near the source; the attribution to groundwater rests on the independent aquifer-side validation in Section S2. The result is observation-consistent evidence that the groundwater pathway is a plausibly first-order contributor to the lower-mainstem PFOS.

S3.1 Hydraulic sensitivity of the coupled mass exchange

The preceding analyses target the in-stream PFOS *prediction*; a separate question is which *hydraulic* parameters govern the surface-water/groundwater *interaction* itself. Because the PFOS plume needs centuries to develop (Section S3; the vadose analysis in the main text), we probe the coupling with the most mobile analyte in the watershed monitoring data, PFHxA (retardation $R \approx 1.6$; weak solid and air–water sorption), which exchanges between aquifer

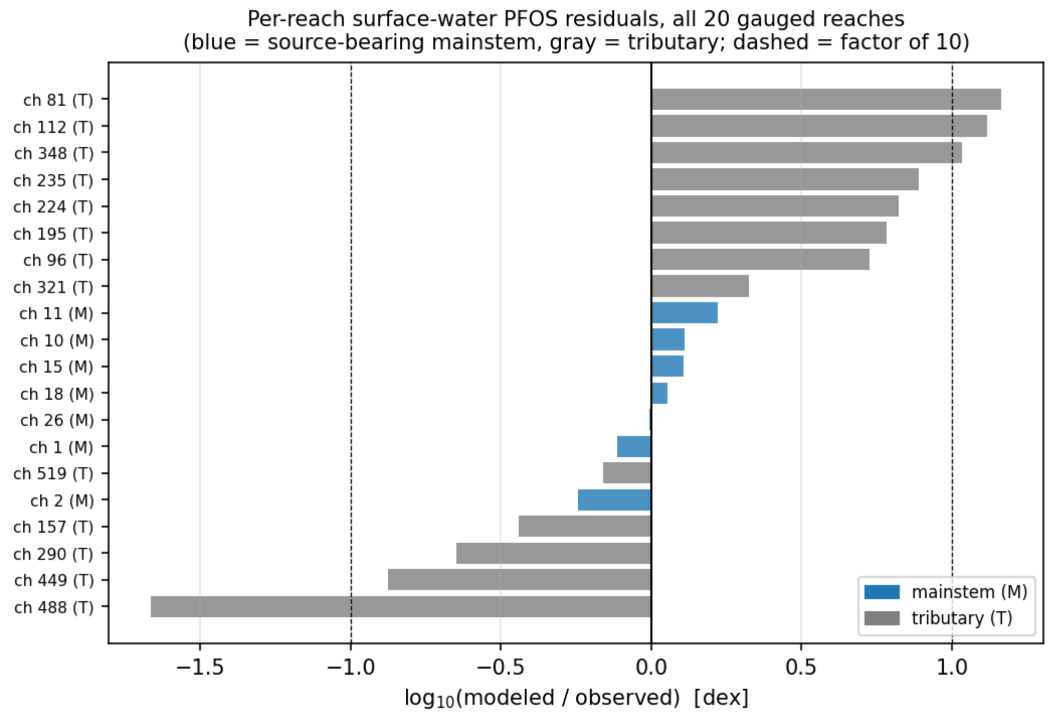

Figure S3: Surface-water PFOS residuals at every one of the 20 gauged reaches, expressed as $\log_{10}$(modeled/observed) (zero is a perfect match; the dashed lines mark a factor of ten). The seven source-bearing mainstem reaches (blue) cluster within ± 0.3 dex — a factor of two — while the tributary reaches (gray) scatter out to a full order of magnitude in both directions. The residual is therefore *structured*, not random: the global parameters constrain the mainstem well, and the unexplained variance lives in local tributary soil heterogeneity that a single basin-wide loading factor cannot resolve.

and stream on a short, tractable window. We run the live coupled model for a 90-day period and perturb each hydraulic parameter one at a time, recording the groundwater–stream water exchange (baseflow) and the groundwater PFHxA discharged to streams (Figure S15).

Two parameters dominate: halving the aquifer hydraulic conductivity cuts both baseflow and PFHxA discharge by 40%, and doubling the streambed conductance raises them by 20% — the two conductances that set baseflow. Recharge, specific yield, vertical conductivity, and dispersivity each change the exchange by under 1% over the seasonal window, because baseflow here is fed from storage rather than from same-season recharge. The PFHxA discharge tracks the baseflow almost exactly (a mobile tracer follows the water), and the transport mass balance closes on every run ($|\text{discrepancy}| \leq 0.01\%$). We note a robustness limit: the most extreme perturbations (doubling aquifer conductivity, halving streambed conductance) drive flow velocities at which the daily transport solve fails to converge — evidence that the coupled solute transport is genuinely sensitive to the hydraulic field, not merely a passive rider on it.

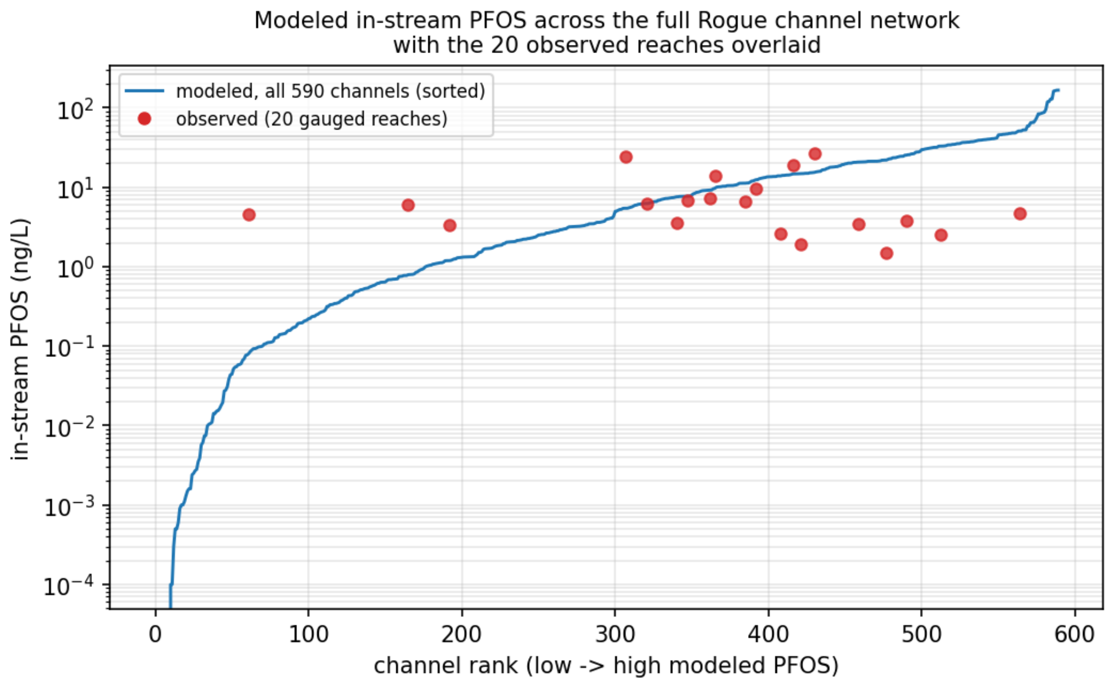

Figure S4: The model's complete in-stream PFOS field. Every one of the 590 channels in the Rogue network is plotted (blue, sorted from lowest to highest modeled concentration), spanning more than four orders of magnitude from headwater channels near zero to the source-influenced lower mainstem. The 20 observed reaches (red) are overlaid at their modeled rank: they fall in the populated upper-middle of the distribution, confirming that the monitoring network samples the concentration range the model resolves rather than its extremes.

S4 Coupled global sensitivity analysis

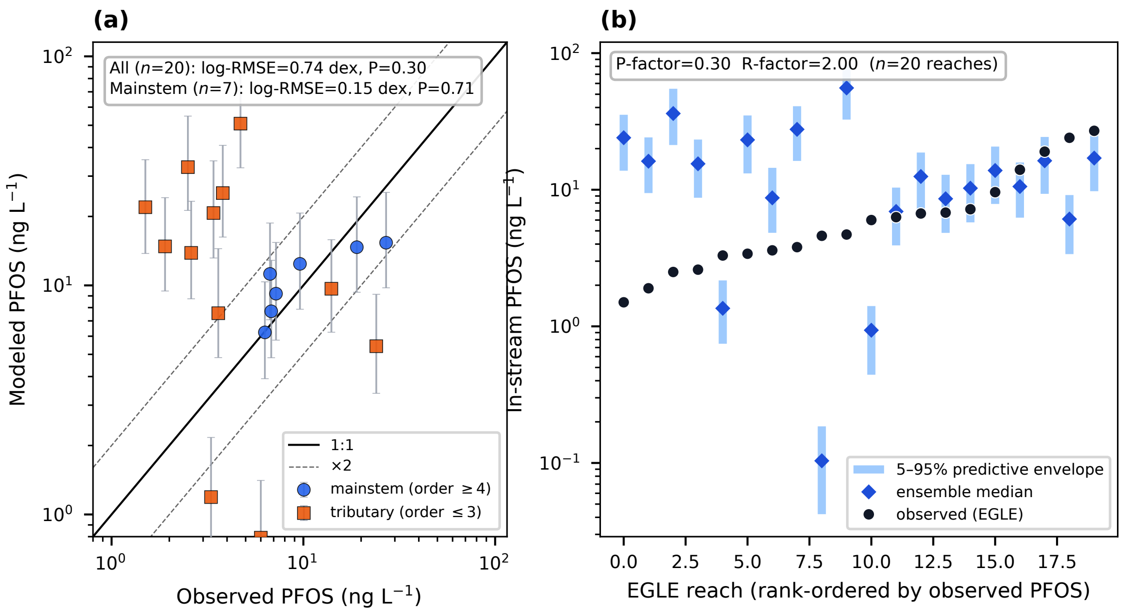

Figure S5: Surface-water predictive uncertainty from the 120-member Latin-hypercube ensemble. (a) Deterministic calibration with the 5th–95th percentile predictive envelope (observed versus modeled per reach, log–log). (b) Predictive uncertainty across reaches: observed concentrations rank-ordered, with the ensemble median and the 5–95% envelope. The source-bearing mainstem reaches lie within the $\times 2$ envelope; the tributaries scatter, consistent with the structured residual of Figure S3.

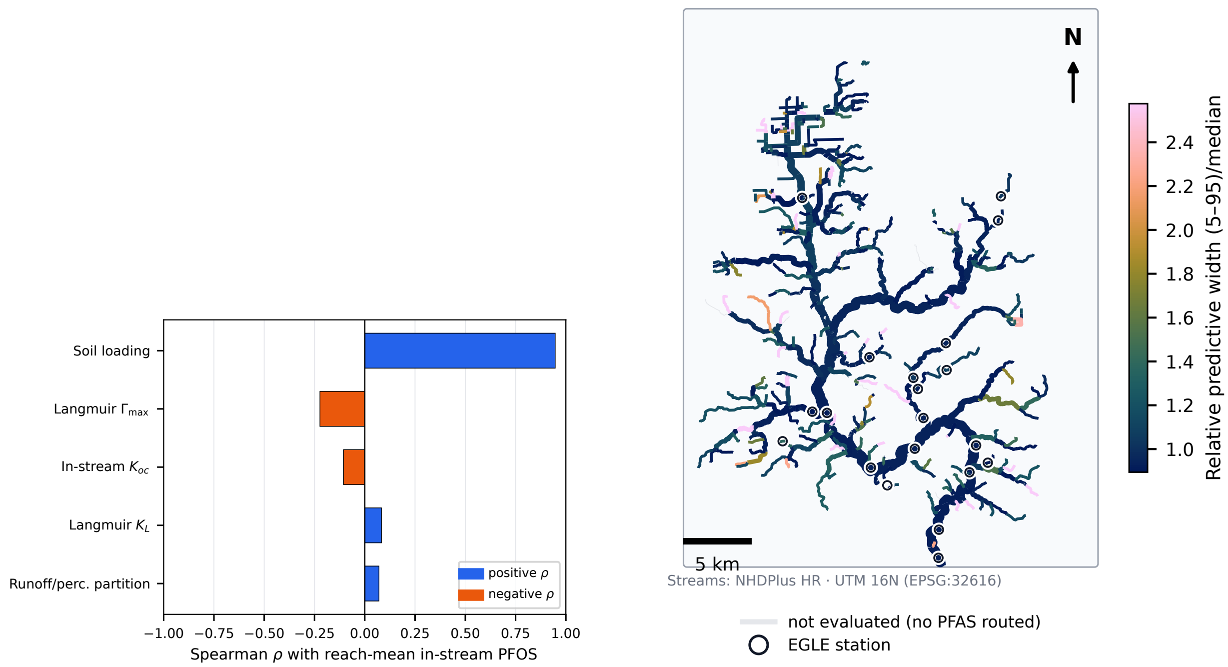

Figure S6: (left) Global parameter influence on the in-stream PFOS prediction (Spearman ρ); the soil-loading multiplier dominates and the in-stream and air–water-interface parameters are nearly inert at the calibrated optimum. (right) Per-reach relative predictive width; the mainstem is more tightly constrained than the headwater tributaries.

Table S1: Per-reach surface-water PFOS at all 20 gauged reaches: observed grab-sample concentration, the calibrated model prediction, and the 5th/95th-percentile bounds of the 120-member uncertainty ensemble (all ng L^{-1}). Class M denotes the seven source-bearing mainstem reaches, T the thirteen tributary reaches; reaches are sorted by observed concentration within each class.

Reach	Class	Obs	Modeled	Env. 5%	Env. 95%
2	M	27.0	15.4	9.8	25.5
1	M	19.0	14.7	9.3	24.3
10	M	9.6	12.4	7.9	20.7
15	M	7.2	9.2	5.8	15.4
18	M	6.8	7.7	4.8	12.9
11	M	6.7	11.2	7.1	18.7
26	M	6.3	6.2	3.9	10.4
290	T	24.0	5.4	3.4	9.1
519	T	14.0	9.7	6.2	15.8
449	T	6.0	0.8	0.4	1.4
348	T	4.7	50.8	32.6	82.4
488	T	4.6	0.1	0.0	0.2
224	T	3.8	25.3	16.3	40.9
321	T	3.6	7.6	4.8	14.5
195	T	3.4	20.7	13.2	35.0
157	T	3.3	1.2	0.7	2.2
96	T	2.6	13.9	8.7	23.3
112	T	2.5	32.8	21.2	54.8
235	T	1.9	14.8	9.5	24.2
81	T	1.5	21.9	13.8	35.4

Table S2: Surface-water PFAS calibration parameters: prior range, calibrated value, and global sensitivity (Spearman rank correlation of the parameter against the in-stream PFOS prediction across the uncertainty ensemble). The soil-loading multiplier carries essentially all of the predictive influence; the in-stream and air–water-interface parameters are weakly identified at this site.

Parameter	Prior range	Calibrated	Spearman ρ
Soil-loading multiplier	[0.05, 0.22]	0.11	+0.95
In-stream K_{oc} multiplier	[0.5, 2]	1	-0.11
Langmuir affinity K_L	[0.07, 0.27]	0.137	+0.08
Langmuir max $\Gamma_{\max}$	[1500, 3500]	2500	-0.22
Runoff/percolation partition	[0.1, 0.4]	0.2	+0.07

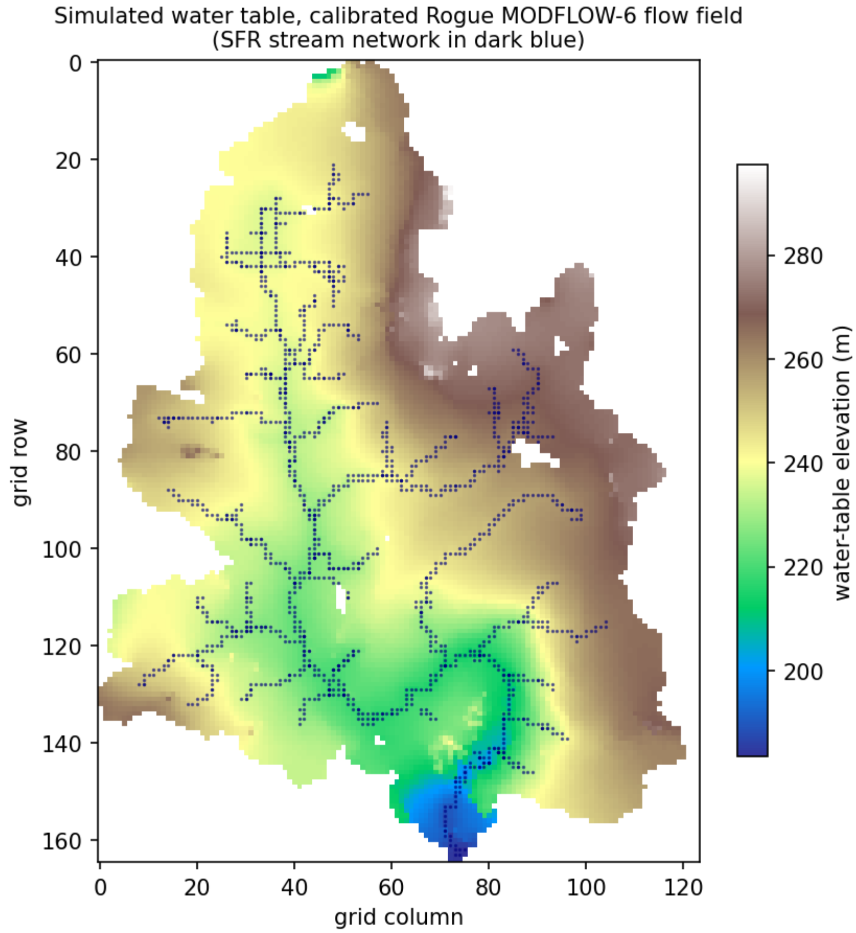

Figure S7: Simulated water-table elevation across the active MODFLOW 6 domain, with the SFR stream network (dark blue) overlaid. The water table mirrors the land-surface topography, descending from the basin divides (∼290 m) toward the dendritic channel network and the basin outlet in the lower domain (∼185 m). The streams sit in the topographic lows, the configuration that produces a gaining (groundwater-discharging) network consistent with the observed baseflow.

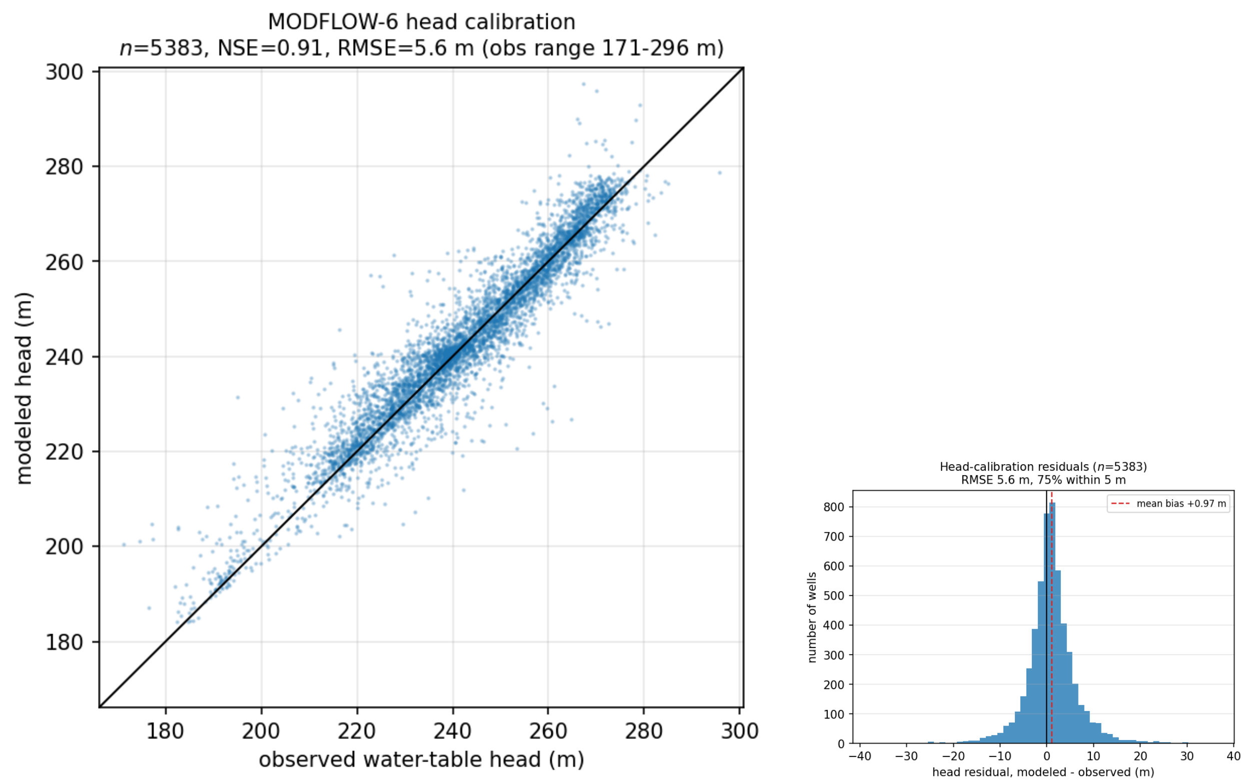

Figure S8: MODFLOW 6 head calibration. (left) Modeled versus observed static water-table elevation at the 5,383 per-cell well observations (raw static-water-level medians, not the kriged surface); NSE 0.91, RMSE 5.6 m. The high NSE is partly carried by the $\sim$125 m topographic range of heads, so the 5.6 m RMSE is the more conservative skill statement. (right) Distribution of the head residuals (modeled $-$ observed): they are nearly symmetric and almost unbiased (mean +1.0 m), with three quarters of wells matched within 5 m, so the conductivity field is not systematically too high or too low across the basin.

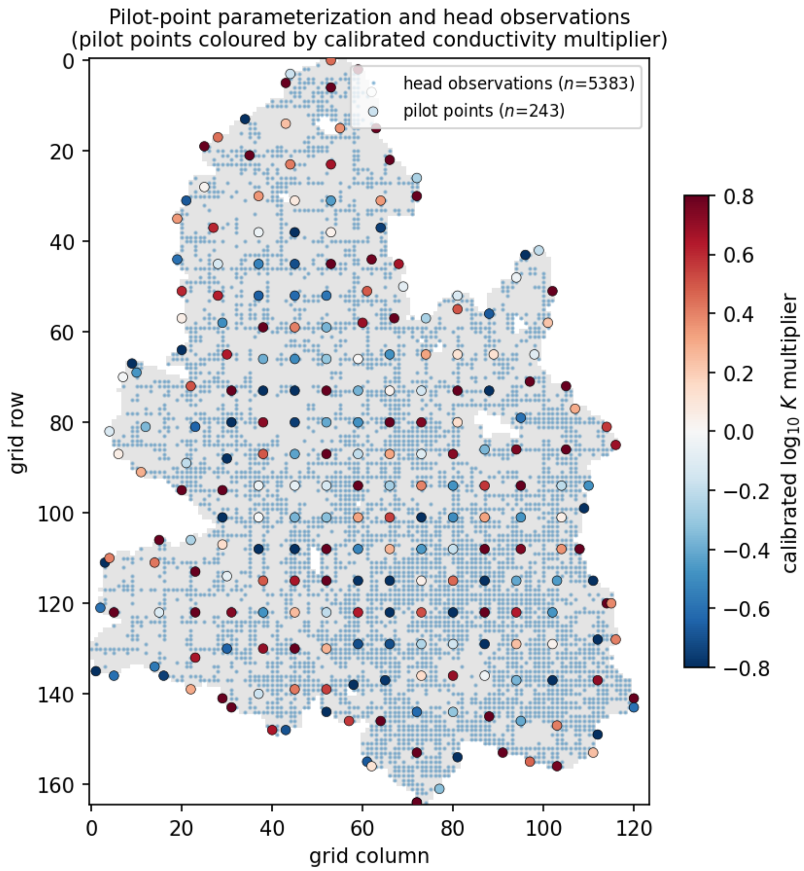

Figure S9: Spatial parameterization of the groundwater flow model. The 243 pilot points (large dots) parameterize the hydraulic-conductivity field by kriging; each is coloured by its calibrated $\log_{10}$ conductivity multiplier (red = enhanced, blue = reduced relative to the kriged prior), so the figure also shows where the inversion adjusted conductivity to fit the heads. The small blue dots are the 5,383 head observation cells that constrain the inversion; their dense, basin-wide coverage is what makes the 243-parameter field identifiable.

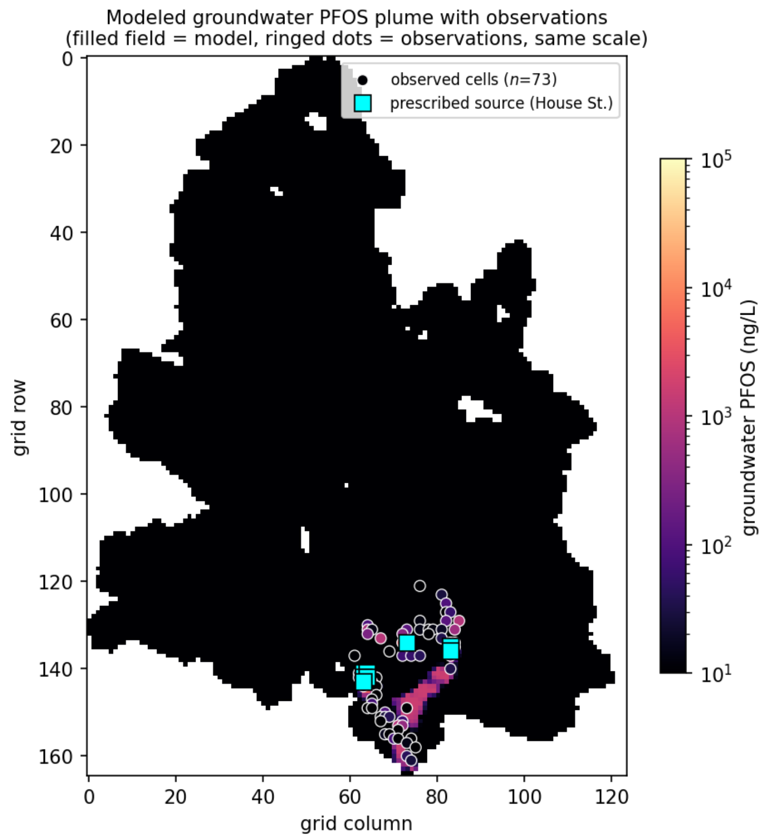

Figure S10: Modeled groundwater PFOS plume in plan view (depth-maximum concentration), with the 73 observation cells (white-ringed dots) and the prescribed House Street source (cyan squares) overlaid on the same logarithmic colour scale. The model develops a high-concentration plume immediately around and downgradient of the prescribed source in the lower basin, where it co-locates with the elevated observations; away from the source the modeled field falls to the ambient background. The elevated observations that lie far from House Street (upper-row dots with no surrounding modeled plume) are the signature of *additional* source areas a single prescribed source does not represent.

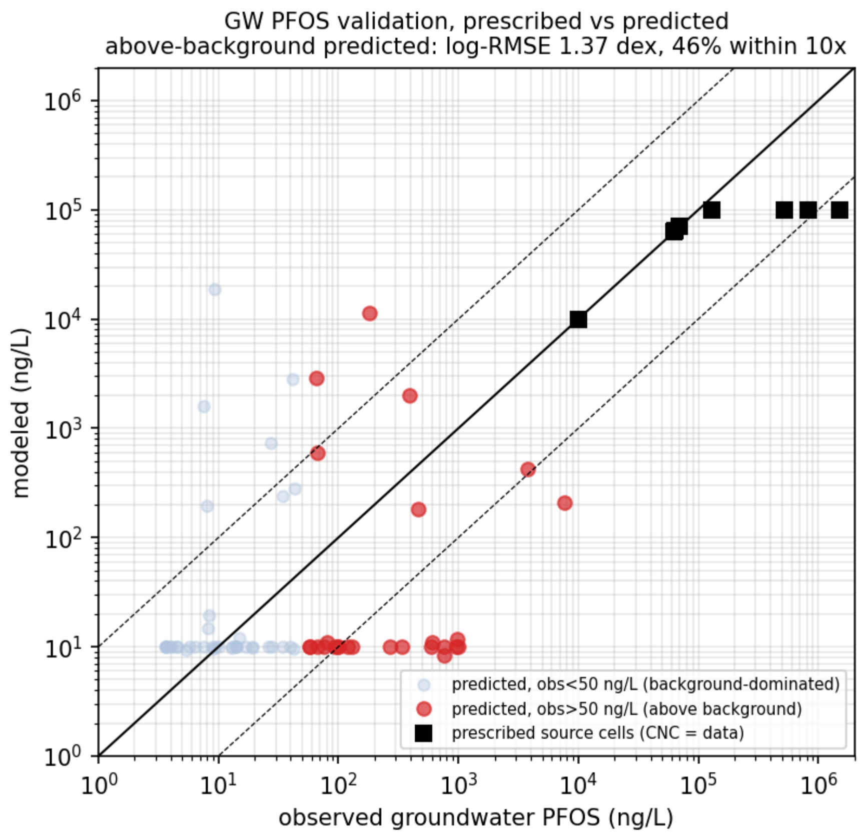

Figure S11: Groundwater PFOS validation, disaggregated to keep the skill statistic honest. *Prescribed* source cells (black squares; concentration set to the data) are not predictions and are excluded from the metric. Among the *predicted* cells, those dominated by the ambient background (observed < 50 ng L^{-1}, light) would inflate any whole-set metric; the honest skill is on the above-background predicted cells (observed > 50 ng L^{-1}, red): log-RMSE 1.37 dex, 46% within a factor of ten. The model reproduces the plume *shape near the source* but predicts ambient background away from it. At the 250 m grid the transport grid Péclet number is ∼25 ($\Delta x/\alpha_L$), so numerical dispersion dominates the prescribed physical dispersivity — an argument for the local grid refinement proposed in the main text.

Table S3: Groundwater PFAS transport parameters and their provenance. "Prescribed" values are set from measurement or the model grid and are not calibrated; the source concentration in particular is anchored to the measured House Street plume rather than fitted, which is why the aquifer-side validation (Figure S11) is an honest test rather than a self-consistency check.

Parameter	Value	Provenance
Aquifer porosity	0.30	literature (glacial outwash/till)
Bulk density	1,800 kg m^{-3}	literature
Freundlich k_f	0.005	PFOS sorption, retardation $R \sim 10$
Freundlich exponent n	0.8	PFOS isotherm (literature)
Longitudinal dispersivity α_L	10 m	prescribed (scale-appropriate)
Transverse dispersivity α_T	1 m	prescribed ($\alpha_L/10$)
Molecular diffusion	10^{-10} m^2 s^{-1}	literature
Ambient background	10 ng L^{-1}	$\sim$median of observations
Source concentration	measured, capped at 10^5 ng L^{-1}	*prescribed* to House St. data
Source-zone threshold	10^4 ng L^{-1}	defines prescribed cells
Advection scheme	TVD	numerical (oscillation control)
Grid resolution	250 m	SWAT$^+$-consistent grid

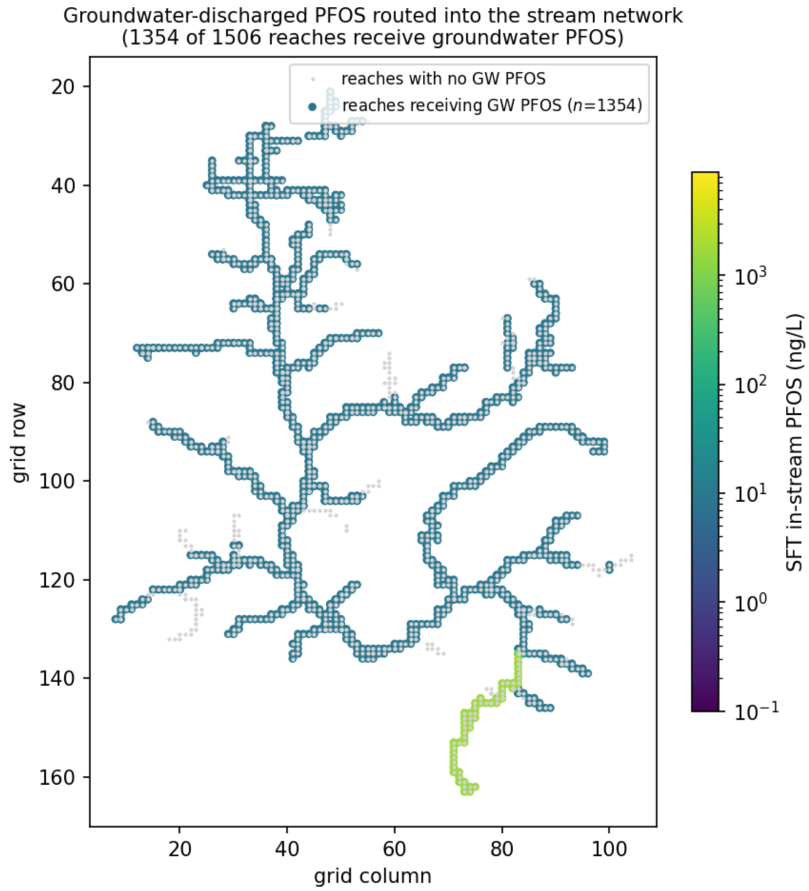

Figure S12: Groundwater-discharged PFOS routed into the stream network by the SFT package. Each stream reach is coloured by the in-stream PFOS concentration delivered by groundwater discharge alone (before dilution by surface streamflow); gray reaches receive none. The groundwater contribution reaches 1,354 of the 1,506 reaches but is heavily concentrated near the contaminated corridor (the bright high-concentration reaches), where it loads the same lower-mainstem channels the surface model under-predicts. This is the spatial mechanism behind the joint-calibration result.

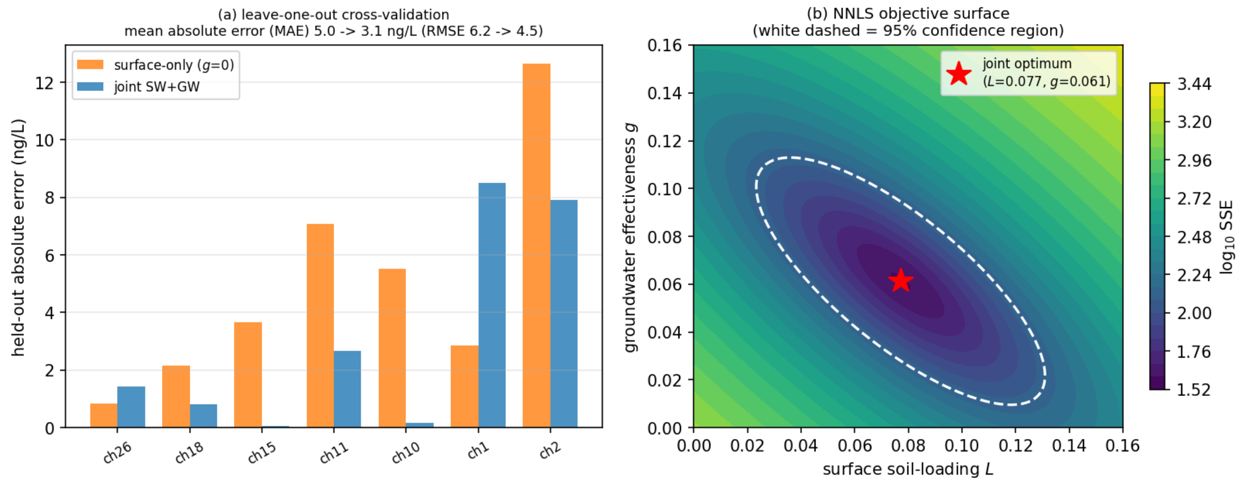

Figure S13: Joint-calibration diagnostics. (a) Leave-one-out cross-validation: at each main-stem reach the model is refitted on the other six reaches and the held-out reach is predicted. The joint surface-plus-groundwater model (blue) beats the surface-only model (orange) at the source-influenced lower reaches and lowers the mean held-out absolute error (MAE 5.0 $\rightarrow$ 3.1 ng L^{-1}; the equivalent held-out RMSE 6.2 $\rightarrow$ 4.5 ng L^{-1} is the figure quoted in the main text and Table 3), so the added groundwater term generalizes rather than merely fitting in-sample. (b) The non-negative-least-squares objective surface in the (L, g) plane; the red star is the joint optimum ($L = 0.077$, $g = 0.061$) and the white dashed contour is the joint 95% confidence region. The region clearly excludes $g = 0$ (the bottom edge), i.e. a surface-only model, and the elongated, tilted shape shows the expected trade-off between the surface loading and the groundwater effectiveness — raising one and lowering the other partly compensates, which is why the two must be fitted together.

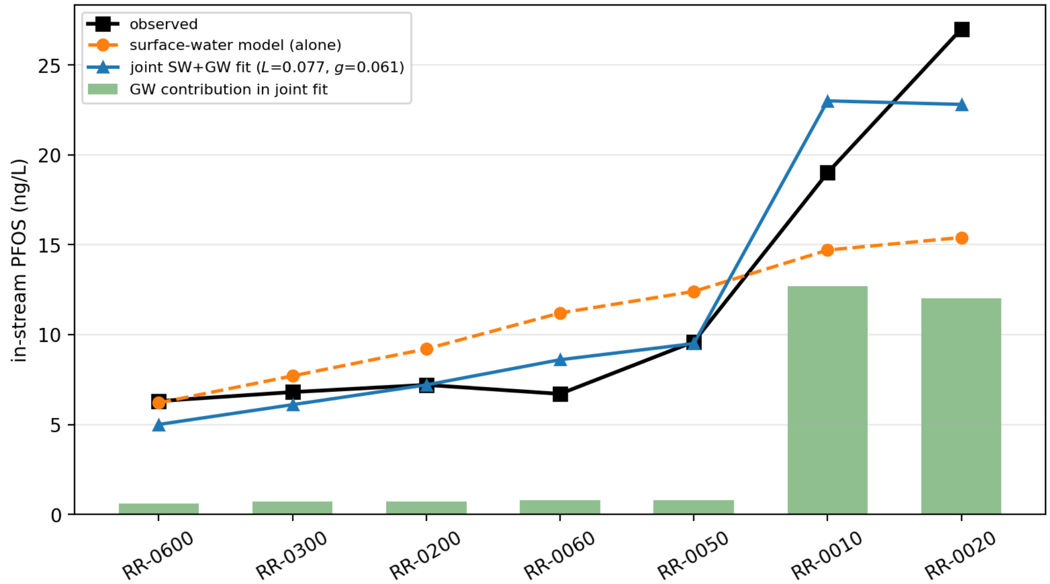

Figure S14: Joint surface-water + groundwater calibration on the Rogue mainstem (upstream to downstream). The joint fit (blue) tracks the observed rise (black) where the surface-water model alone (orange) under-predicts; the fitted groundwater contribution (green bars) is small in the headwaters and large at the lower mainstem near the Wolverine source, closing the lower-mainstem residual (g = 0.061; joint 0.07 versus surface-only 0.15 dex).

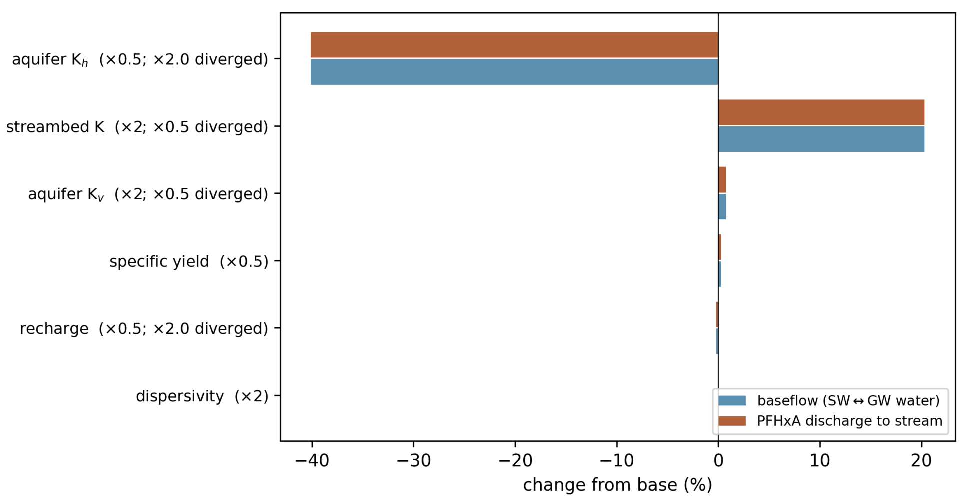

Figure S15: One-at-a-time hydraulic sensitivity of the coupled model (mobile tracer PFHxA, 90-day run): percent change from baseline in the groundwater–stream water exchange (baseflow) and in the PFHxA mass discharged to streams, for each hydraulic parameter (perturbation factor in parentheses). Aquifer K_h and streambed conductance dominate; recharge, storage, K_v, and dispersivity are negligible on the seasonal timescale. Factors marked “diverged” drove the daily transport solve to non-convergence at high flow velocity.

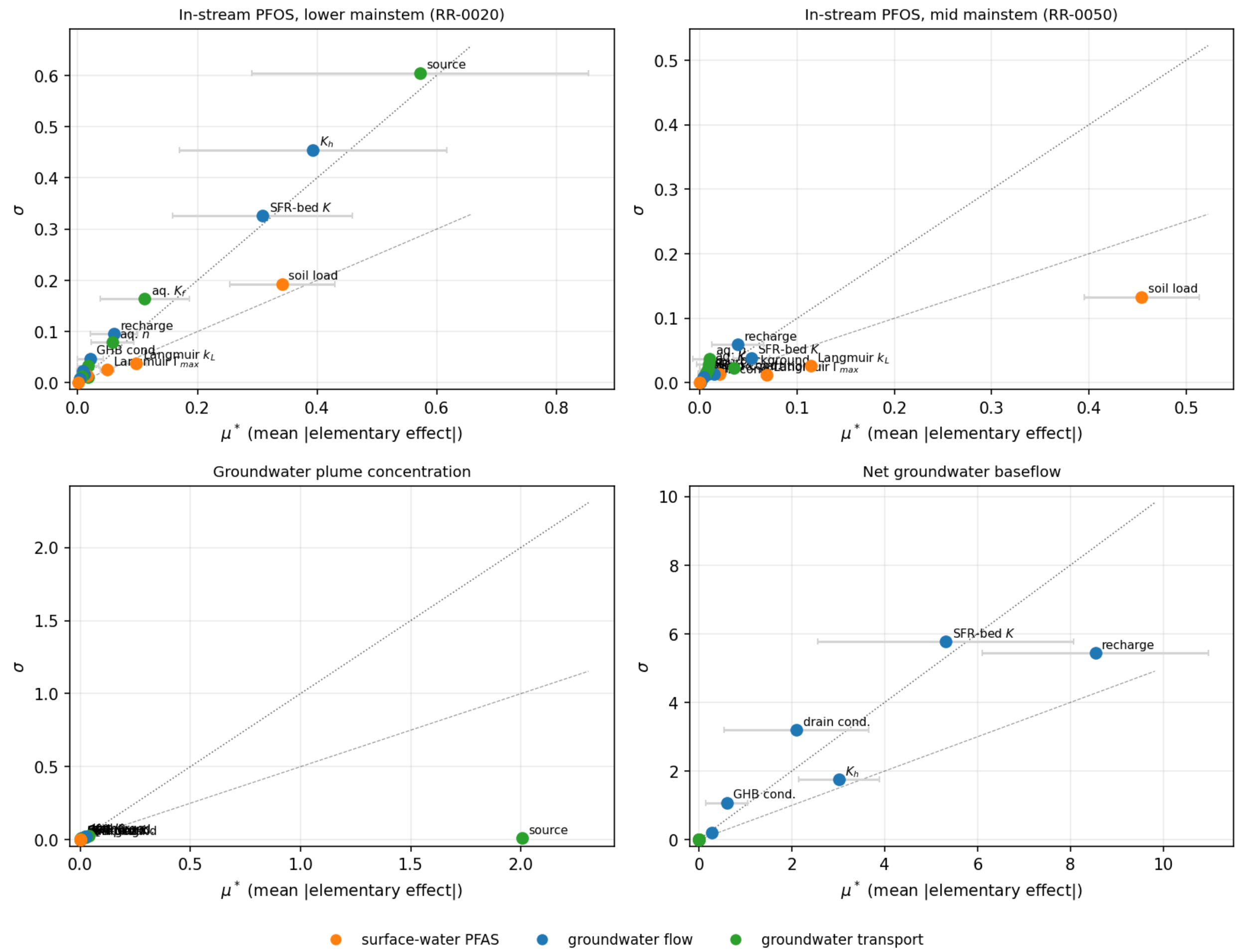

Figure S16: Coupled Morris elementary-effects screening of all 18 surface-water and groundwater parameters, one panel per quantity of interest. Each point is a parameter, positioned by its mean absolute elementary effect μ^* (importance, horizontal) and the spread σ of its effects (interaction and nonlinearity, vertical); horizontal bars are bootstrap 95% confidence intervals on μ^*. Colour denotes the model leg (surface-water PFAS, groundwater flow, groundwater transport). Concentration quantities are analysed in $\log_{10}$ space. The dotted line marks $\sigma = \mu^*$. The in-stream PFOS at the lower mainstem is governed by the groundwater source and aquifer conductivity (with strong interactions, $\sigma \approx \mu^*$) and secondarily by the surface soil-loading; at the mid mainstem the surface soil-loading dominates (near-linear, $\sigma \approx 0.5\,\mu^*$); the groundwater plume is governed by the transport parameters and the baseflow by the flow parameters. Design: $r = 20$ trajectories, 380 coupled evaluations on a 64-core cloud instance; 13% of samples (near-stalled groundwater corners) did not complete and are handled by per-step elementary-effect computation. Full ranking in the main-text sensitivity table.